Exercise 0: Rewind – Machine Learning Basics

Task 1: Machine Learning Problems

Identify the general type of machine learning tasks listed below (classification, regression, clustering). For each of these, name a model / algorithm you could use to solve the task, as well as a metric for evaluation.

- House prices: Given past sales data, determine the expected price of a newly listed property.
- Banking transactions: Based on past banking data, determine whether a new transaction is suspicious.
- Buying habits: Analyze purchase histories to discover groups of people with comparable buying habits.
- Movie reviews: Analyze movie reviews and predict the expected star rating (1 to 5).

Task 2: Model Evaluation

- Why is it necessary to split data into training and test sets instead of training and testing on the complete dataset?
- What is the purpose of a separate validation set?
- Name three metrics you could use for evaluating a model trained on a classification task. For each of them, describe a situation in which that metric is especially useful and explain a problem, why you can't solely rely on that metric in general.

Task 3: Improving Weak Models

The neural network you trained on a regression task yields very low error on the training data, but after the training is finished, the model's predictions still result in high error on the test data.

- What is this phenomenon called?
- What possible reasons could cause this problem?
- What methods could you apply to address it?

Task 4: House Price Prediction (Python)

Check out the *housing_prices.csv* file from Moodle. It contains information about houses, as well as their sale prices. Your task is to train a machine learning model that predicts the house prices based on the other features in the dataset.

- Choose an appropriate model for the task and give your reasoning why.
- Use all available features for making predictions. For each of these features, explain how you encoded / scaled them for your model or explain why encoding / scaling isn't necessary.
- Choose 2 metrics for evaluating your model and explain your choice.
- Train and test your model using a 70/30 split and interpret the results.
- *Optional:* Train other model architectures on the same data and compare their performance.